DESIGN AND FABRICATION OF SIGN LANGUAGE TRANSLATION GLOVES

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ABSTRACT

- This project introduces a smart glove-based translation system designed to bridge the communication gap faced by individuals with hearing and speech impairments. The glove uses flex sensors embedded in its fabric to detect finger movements, which are then processed by an Arduino and microcontroller to interpret sign language gestures. These gestures are translated into real-time text on an external screen.

- The glove is designed to be lightweight, cost-effective, and user-friendly, promoting widespread adoption and accessibility. By offering a portable and affordable solution, this project takes a significant step toward integrating assistive technology into everyday communication. It aims to enhance the quality of life for individuals with communication impairments by enabling real-time and accurate sign language translation

OBJECTIVE

- This project aims to offer a more interactive and real-time solution for the hearing and speech-impaired by translating sign language into text using a smart glove system with basic C++ programming.

- It provides a cost-effective, portable, and independent communication tool for users.

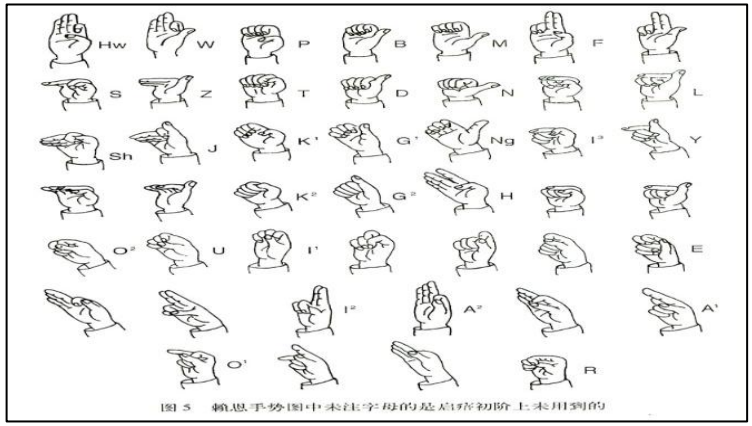
- This project utilizes machine learning to enhance gesture recognition accuracy, resulting in a more reliable and adaptable translation system.

EXISTING TECHNOLOGY

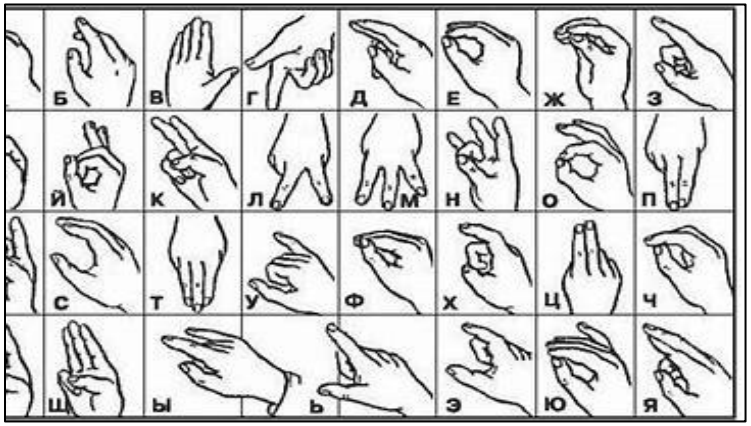
Most existing translation gloves:

- Are designed for Chinese Sign Language (CSL).
- Are designed for Japanese Sign Language (JSL).
- Translate only a limited number of words or alphabets.
- Rely on bulky components or are not wearable for daily use.
- Lack of support for Indian Sign Language (ISL):ISL has unique signs that differ significantly from ASL.
- Very few devices are available for real-time ISL translation.

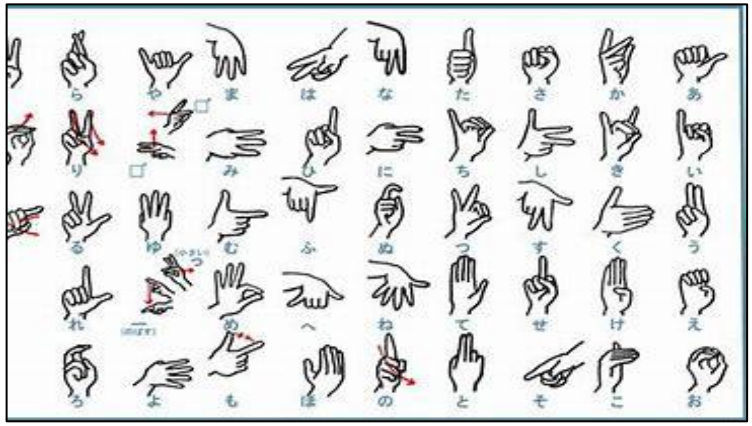
EXISTING TECHNOLOGY



Chinese Sign Language

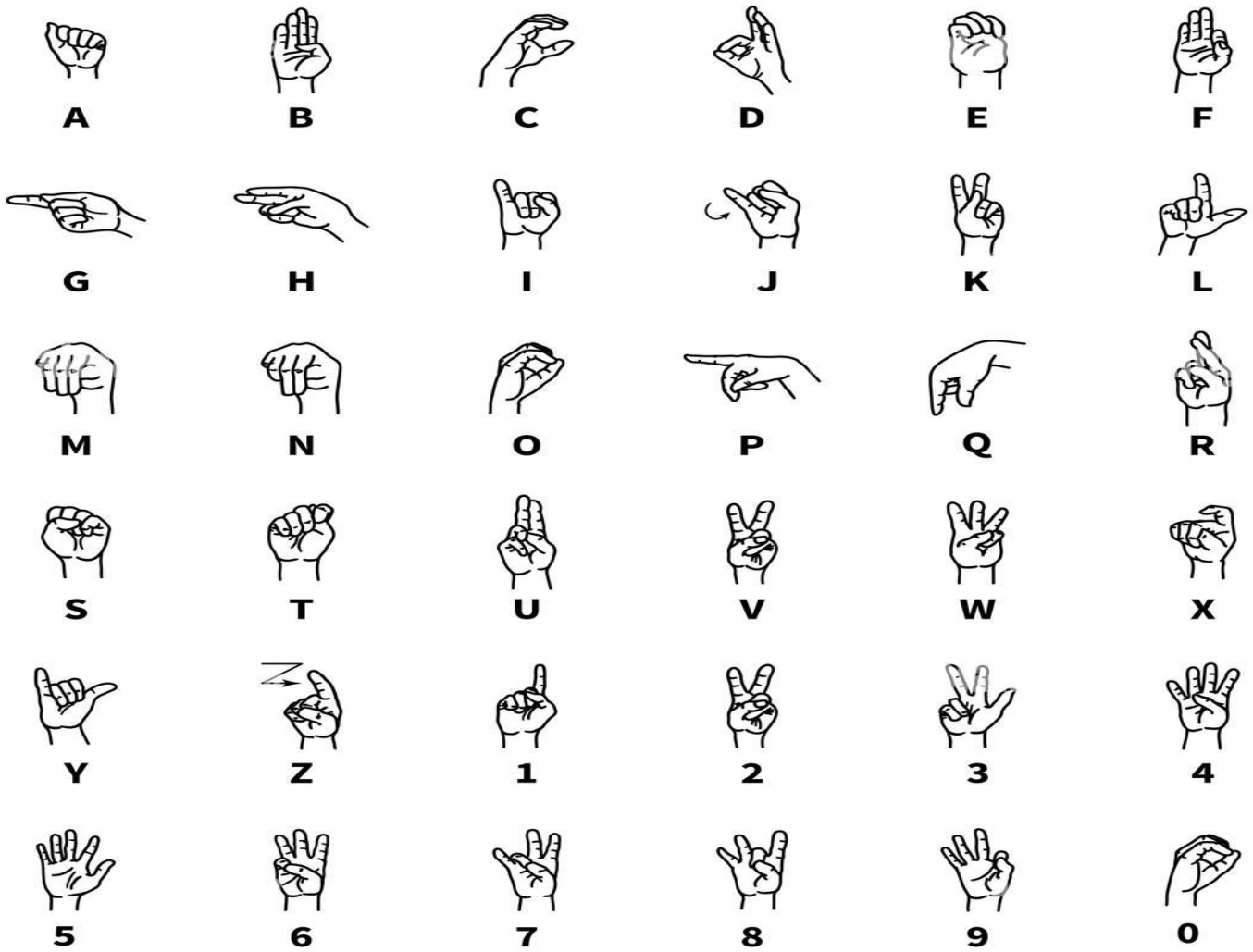


Ukrainian Sign Language



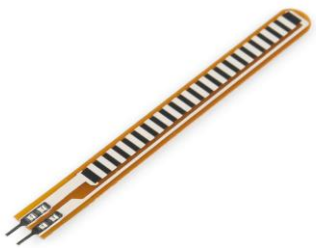
Japanese Sign Language

Standard Sign

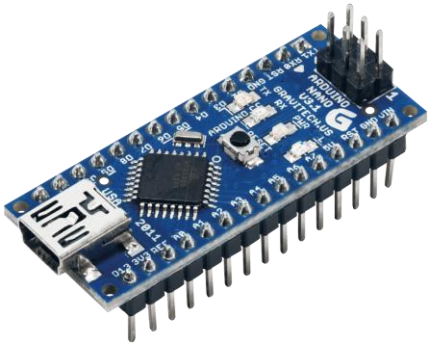


Indian Sign Language (ISL)

COMPONENTS USED



- A **flex sensor** is a type of sensor that measures how much it bends or flexes. It’s usually a thin strip that changes its electrical resistance when you bend it.



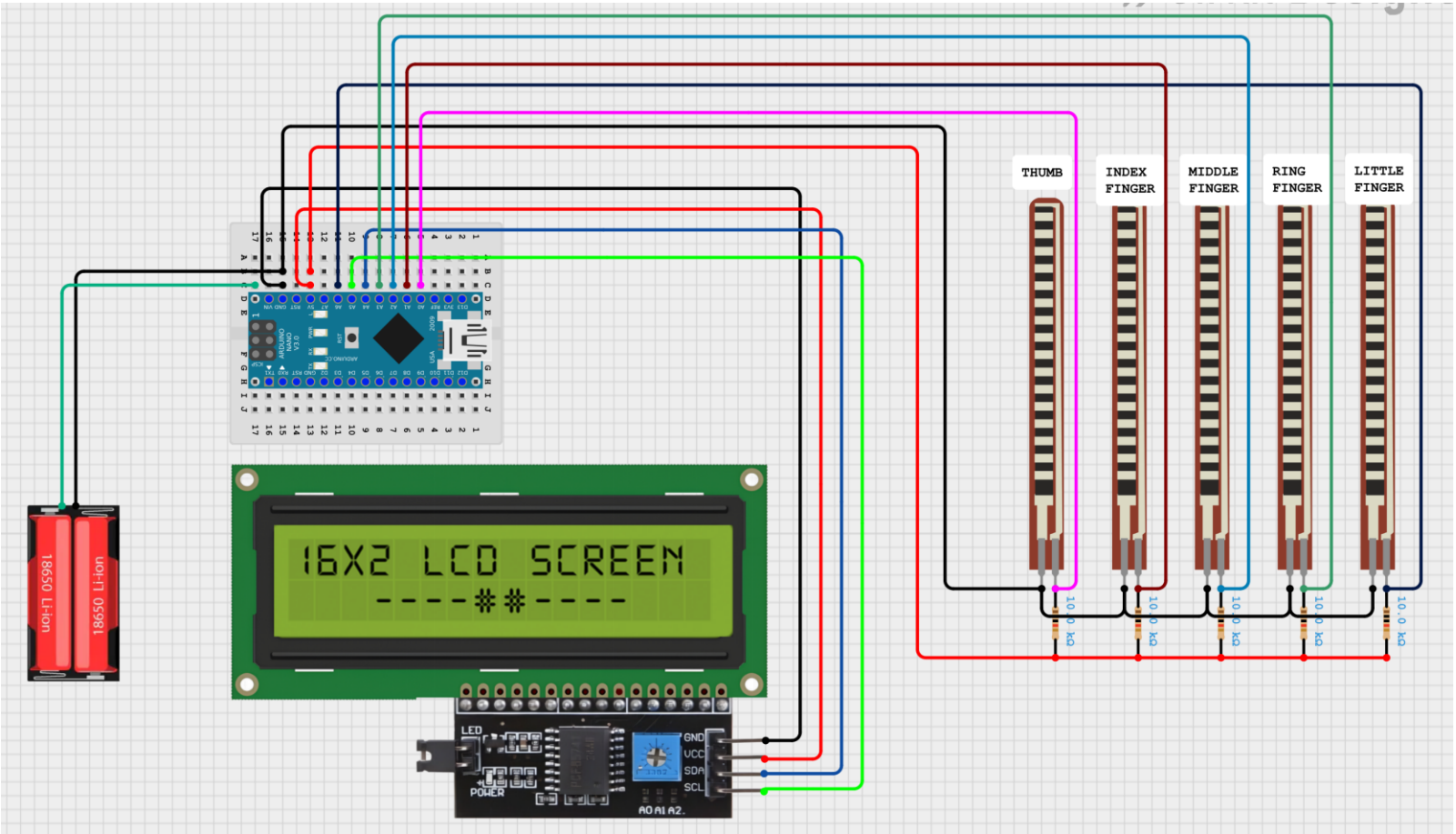
- **Arduino** is a popular microcontroller platform that's super beginner-friendly. It includes both hardware (the board) and software (Arduino IDE) to write and upload code.

- A **lithium battery** is a lightweight, rechargeable battery that provides portable power for electronic devices.

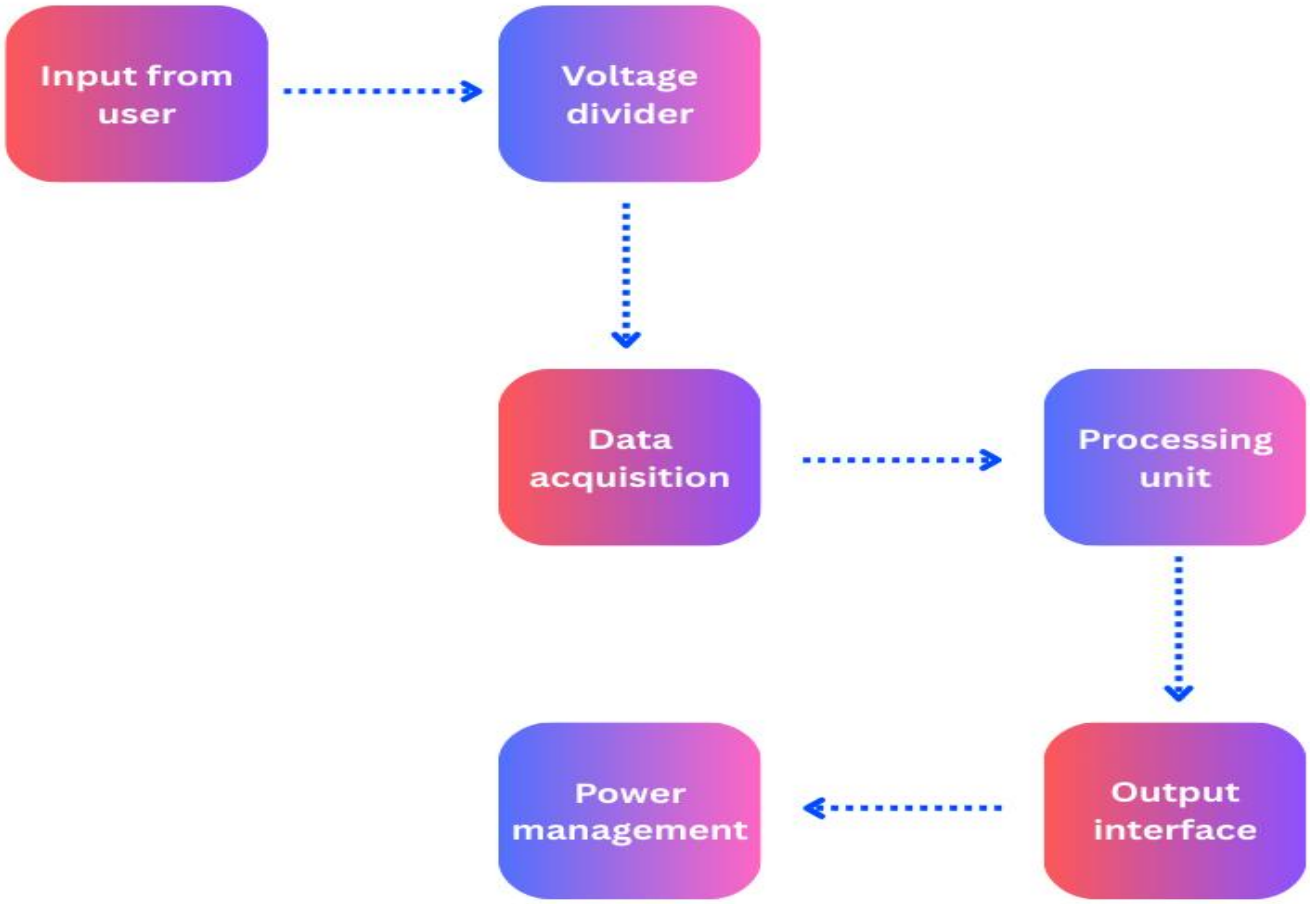
- It’s a screen that shows text, numbers, or basic symbols. In Arduino projects, the most commonly used type is the **16x2 LCD**, which can show 2 lines with 16 characters each.

- In your project, the glove is the foundation that holds all the components like flex sensors, wires, and possibly other sensors.

CIRCUIT DIAGRAM



BLOCK DIAGRAM



WORKING

Input Subsystem

Flex Sensors (x5): -- Detect bending of each finger (Thumb, Index, Middle, Ring, Little).

-- Analog voltage output proportional to bend angle.

Voltage Divider Circuit: -- 10kΩ resistors used to convert variable resistance into readable voltage.

Data Acquisition

Arduino Nano (Microcontroller): -- Reads analog inputs from flex sensors (A0–A4).

-- Converts analog signals to digital using built-in ADC.

-- Performs thresholding and logic mapping of gesture patterns.

Processing Unit

Gesture Recognition Algorithm: -- If-else or switch-case logic to match sensor values with ISL gestures.

-- Can be upgraded to use a decision tree or neural network model.

Output Subsystem

16x2 LCD (I2C Interface): -- Displays recognized letter, word, or phrase

-- Communicates via SDA/SCL pins

Power Management

18650 Li-Ion Battery Pack: -- Provides 7.4V regulated down to 5V for Arduino.

-- TP4056 charger circuit for recharging

ADVANTAGES & APPLICATIONS

Advantages:

- Real-time ISL gesture recognition.
- Affordable and easy to build.
- Portable and battery-operated.
- Expandable to more gestures or languages.

Applications:

- Communication aid for deaf and dumb individuals.
- Educational tool for learning ISL.
- Use in hospitals, banks, and public services to serve differently-abled individuals.
- Potential use in voice-controlled systems for people with speech impairment.

ESTIMATION OF COST

COMPONENTS	COST(Rs)
FLEX SENSOR (5 NOS)	1300
ARDUINO NANO	200
BATTERY PACK	120
LCD DISPLAY WITH I2C MODULE	170
10K OHM RESISTOR	10
CONNECTING WIRES	70
UNIVERSAL PCB BOARD	60
TOTAL COST	1930

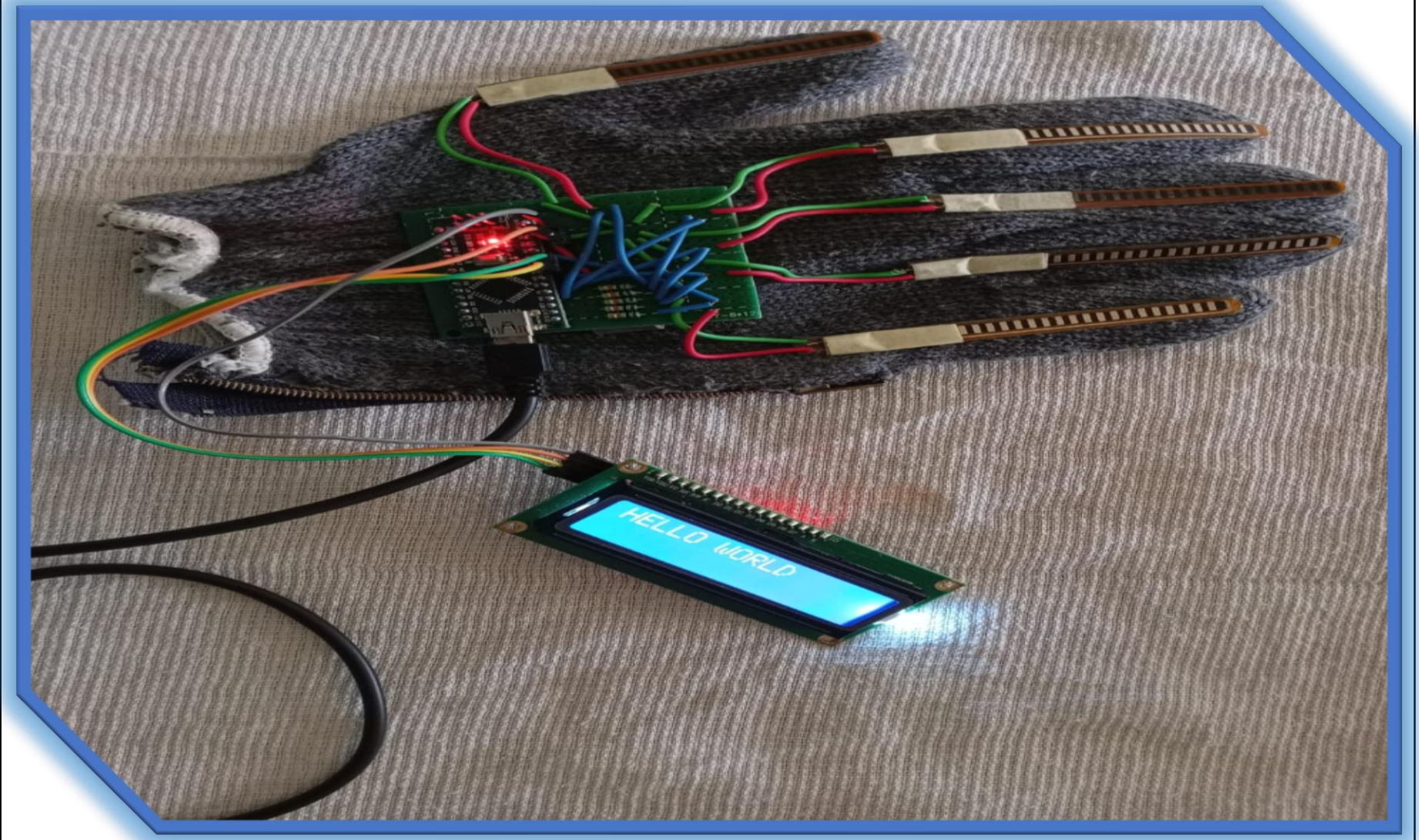
FUTURE ENHANCEMENTS

- Integration of machine learning algorithms to improve gesture accuracy.
- Development of a mobile app for real-time translation, voice output, and data logging.
- Support for dynamic gestures (moving signs), not just static ones.
- Addition of speech-to-sign translation to assist communication in both directions.
- Use of flexible PCBs and washable sensors for a commercial glove design.

CONCLUSION

- This project provides a cost-effective, real-time translation glove for Indian Sign Language.
- Helps bridge the communication gap between deaf/mute individuals and the general public.
- It’s a wearable, lightweight, and expandable prototype.
- The glove opens doors for future commercial and assistive technology development in India.

FINAL PROTOTYPE



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- Lee BG, Lee SM. Smart wearable hand device for sign language interpretation system with sensors fusion. IEEE Sensors Journal. 2017 Dec 4;18(3):1224-32.

THANK
YOU!